

Indrajeet Mondal

MCA Student | Full-Stack & ML Developer
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ABOUT ME

A fast-learning MCA student with a strong ability to adapt to new technologies and tech stacks quickly. Proficient in leveraging AI and LLMs to scaffold robust foundation architectures, then customising and fine-tuning them to meet specific project requirements — significantly accelerating development without compromising quality. Possesses solid programming fundamentals and a practical, outcome-driven approach to problem-solving, with a keen interest in building intelligent, full-stack applications.

EDUCATION

Master of Computer Applications (MCA)

Narula Institute of Technology, Kolkata • SGPA: 8.32 | 8.64 | 8.84 | Sem 4 Awaited

B.Sc. in Computer Science (Honours)

St. Xavier's College, Kolkata • CGPA: 6.11 / 10

Class X & XII — CBSE

Air Force Golden Jubilee Institute • 94.6% (both boards)

TECHNICAL SKILLS

Languages & Web: Python, JavaScript, SQL, HTML/CSS, React.js, Tailwind CSS

Backend & DB: Node.js, Express.js, Prisma ORM, PostgreSQL

AI / ML: LangChain, Groq API, face-api.js, Scikit-learn, XGBoost, Pandas, NumPy

Tools: Git, GitHub, Streamlit, VS Code, Postman, Vercel

PROJECTS

Face Detector — Real-Time Face Detection Web App

React • face-api.js • Tailwind CSS • Express.js • Prisma • PostgreSQL

- Built a full-stack real-time face detection app using face-api.js for in-browser ML inference with a React + Tailwind CSS frontend.
- Developed a RESTful Express.js backend with Prisma ORM and PostgreSQL to persist user sessions and detection history.

RAG Document Assistant — AI-Powered PDF Q&A Tool

Python • LangChain • Groq API • Streamlit

- Built a RAG pipeline enabling users to upload PDFs and query content in natural language using LangChain and Groq API.
- Deployed as a Streamlit web app with real-time Q&A; handled chunking, embedding, and retrieval end-to-end.

Crop Yield Prediction — Comparative ML Study

Python • Scikit-learn • XGBoost • Pandas • Git (4-branch methodology)

- Compared imputation strategies (mean, mode, median) across 4 Git branches; XGBoost and Random Forest achieved the best accuracy while distance-based models performed poorly.
- Documented findings in a research paper on how imputation method choice impacts ML model performance in agricultural datasets.

PUBLICATIONS

Robustness of Machine Learning Models in Agricultural Yield Prediction Across Statistical Data Preprocessing Techniques

Indrajeet Mondal, Sumana Das, Suseta Datta, Debrupa Pal, Subhasree Bhattacharjee

2025 2nd International Conference on Emerging Trends in Electronic Devices and Computational Techniques (EDCT) • IEEE • DOI: 10.1109/EDCT65302.2025.11496342

- Studied the effect of different imputation strategies (mean, median, mode, null removal) on ML model performance for agricultural crop yield prediction.
- Demonstrated that tree-based ensemble models (Random Forest, Gradient Boosting) achieved $R^2 \approx 0.95$, significantly outperforming linear models, and that model robustness was consistent across all preprocessing pipelines.